CS5340 - Uncertainty Modelling in AI

Lecture 9: Monte Carlo Inference (Sampling)

Scribed by: Claude (Anthropic) | Curated By: Assoc. Prof. Lee Gim Hee

Semester 1, AY2026/27 — National University of Singapore

Disclaimer: This scribe was generated by **Claude (Anthropic)** and curated by **Assoc. Prof. Lee Gim Hee**. It is a **supplementary revision aid** that summarizes the lecture slides, which remain the **primary learning material** prepared by the lecturer. Despite curation, the scribe may still contain errors, omissions, or simplifications. Please use it only alongside the official slides, textbook references, and your own lecture notes.

1 Lecture Overview

- **Motivation:** the exact inference algorithms of Lectures 4–5 (variable elimination, belief propagation, junction tree) and the exact E-steps of Lectures 7–8 all rely on sums or integrals that we can evaluate in closed form. In large, loopy or continuous models these **integrals become intractable**. **Monte Carlo** methods replace each intractable integral by an **average over random samples**.
- **What carries over:** the sampling algorithms reuse the factorization of DGMs (Lecture 2) for **ancestral** and **importance** sampling, the **Markov blanket** (Lectures 2–3) for **Gibbs** sampling, and the EM algorithm (Lecture 7) for **Monte Carlo EM**. The Markov chain machinery reappears from the HMM (Lecture 8), now as a tool rather than a model.
- **Roadmap:**
  1. **The Monte Carlo principle**: sample averages converge to expectations (law of large numbers), with error $O(1/\sqrt{N})$ (central limit theorem).
  2. **Independent-sample methods**: rejection sampling, importance sampling, ancestral sampling; use in Monte Carlo EM.
  3. **Markov chain Monte Carlo (MCMC)**: the Metropolis-Hastings algorithm, burn-in, and why it works (stationarity via detailed balance; convergence via irreducibility and aperiodicity).
  4. **Special cases of MH**: the Metropolis algorithm (symmetric proposal) and Gibbs sampling (full-conditional proposal, acceptance always 1).
  5. **Sampling from continuous distributions** with the inverse CDF.

Learning Outcomes

By the end of this lecture, you should be able to:

1. Explain the **Monte Carlo principle** and its justification for sampling methods.
2. Apply **Rejection**, **Importance**, **Ancestral**, **Metropolis-Hastings**, **Metropolis** and **Gibbs** sampling methods to do maximal probability, approximate inference, and expectation.
3. Use **Markov chain properties**, i.e. homogeneous, stationary distribution, irreducibility, aperiodicity, ergodicity and detailed balance, to show validity of the MH algorithm.

2 History and Motivation

2.1 A Short History

- The **Metropolis algorithm** was selected as one of the **top 10 algorithms** with the greatest influence on science and engineering in the 20th century [Beichl & Sullivan, 2000].
- The Monte Carlo method was invented by **Stanislaw Ulam** (1909–1984) in 1946 while **playing solitaire** during convalescence from an illness. He wanted to **compute the chance** that a solitaire laid out with 52 cards would come out successfully. Exhaustive combinatorial calculation proved hopeless, so he instead laid out several solitaires **at random** and **counted** the number of successful plays.
- **Pioneers:** Stanislaw Ulam, John von Neumann, Nicholas Metropolis, Marshall Rosenbluth, Edward Teller and Augusta H. Teller (the last five are the authors of the 1953 paper introducing the Metropolis algorithm, together with Arianna Rosenbluth).

Key Rule: The Idea Behind Monte Carlo Sampling

Ulam's idea of **selecting a statistical sample to approximate a hard combinatorial problem by a much simpler problem** is at the heart of modern Monte Carlo simulation: replace an exact (intractable) count, sum or integral by a **frequency** or **average** over random draws.

2.2 Why Do We Need Sampling?

Given unknown variables $X \in \mathcal{X}$ and data $Y \in \mathcal{Y}$, the following typically **intractable integration problems** are central to Bayesian statistics.

Definition : Three Intractable Integrals of Bayesian Inference and Learning

1. **Normalization.** To obtain the posterior from the prior $p(x)$ and the likelihood $p(y \mid x)$, the **normalizing factor** in Bayes' theorem must be computed:
$$p(x \mid y) = \frac{p(y \mid x)\, p(x)}{\int_{\mathcal{X}} p(y \mid x')\, p(x')\, dx'}.$$
2. **Marginalization.** Given the joint posterior of $(X, Z) \in \mathcal{X} \times \mathcal{Z}$, we often want the **marginal posterior**
$$p(x \mid y) = \int_{\mathcal{Z}} p(x, z \mid y)\, dz.$$
3. **Expectation.** The objective of the analysis is often a **summary statistic**
$$\mathbb{E}_{p(x|y)}\big(f(x)\big) = \int_{\mathcal{X}} f(x)\, p(x \mid y)\, dx$$
for some function of interest $f : \mathcal{X} \to \mathbb{R}^{n_f}$ integrable w.r.t. $p(x \mid y)$ (e.g. the posterior mean, $f(x) = x$, or a posterior probability, $f(x) = \mathbb{1}[x \in S]$).

Each integral is over a **large-dimensional space**, so it is intractable to compute exactly (for a discrete space, it is a sum with exponentially many terms – cf. the K^N paths of an HMM in Lecture 8).

Definition : Optimization

- The goal of optimization is to extract the solution that **minimizes some objective function** from a large set of feasible solutions.
- This set can be **continuous and unbounded**.
- In general, it is **too computationally expensive** to compare all the solutions to find out which one is optimal. Sampling lets us explore the high-value regions only.

3 The Monte Carlo Principle

Definition : Monte Carlo Approximation of a Density

Draw an **i.i.d. set of samples** $\{x^{(i)}\}_{i=1}^{N}$ from a target density $p(x)$ defined on a **high-dimensional space** $\mathcal{X}$ (the set of possible configurations of a system, the space on which the posterior is defined, or the combinatorial set of feasible solutions). These N samples **approximate the target density** by the **empirical point-mass function**

$$p_N(x) = \frac{1}{N}\sum_{i=1}^{N} \delta_{x^{(i)}}(x),$$

where $\delta_{x^{(i)}}(x)$ denotes the **delta-Dirac mass** located at $x^{(i)}$.

3.1 Laws of Large Numbers

Definition : Weak and Strong Laws of Large Numbers

Let $X_1, X_2, \ldots$ be i.i.d. with mean $\mu = \int X\, p(X)\, dX$, and $\overline{X}_n = \frac{1}{n}\sum_{i=1}^{n} X_i$.

- **Weak law: $\overline{X}_n$ converges in probability** to μ, $\overline{X}_n \xrightarrow{P} \mu$ as $n \to \infty$. That is, for any $\varepsilon > 0$,
$$\lim_{n\to\infty} \Pr\left(|\overline{X}_n - \mu| > \varepsilon\right) = 0.$$
- **Strong law: $\overline{X}_n$ converges almost surely** to μ, $\overline{X}_n \xrightarrow{\text{a.s.}} \mu$ as $n \to \infty$. That is,
$$\Pr\left(\lim_{n\to\infty} \overline{X}_n = \mu\right) = 1.$$

Scribe's note: almost-sure convergence implies convergence in probability, so the strong law is the stronger statement. The weak law says that for each *fixed* large n, a big deviation is unlikely; the strong law says that, with probability one, the *whole sequence* of running averages eventually settles at μ.

3.2 Monte Carlo Integration

Key Rule: The Monte Carlo Estimator

We approximate the integral (or very large sum) $I(f)$ by the **tractable sum** $I_N(f)$:

$$\underbrace{I_N(f) = \frac{1}{N}\sum_{i=1}^{N} f\big(x^{(i)}\big)}_{\text{plays the role of } \overline{X}} \xrightarrow[N\to\infty]{\text{a.s.}} \underbrace{I(f) = \int_{\mathcal{X}} f(x)\, p(x)\, dx,}_{\text{plays the role of } \mu} \qquad x^{(i)} \overset{\text{i.i.d.}}{\sim} p(x).$$

The estimate $I_N(f)$ is **unbiased** and, by the **strong law of large numbers** (applied to the i.i.d. variables $f(x^{(i)})$), it **almost surely converges** to $I(f)$.

Key Rule: Variance of the Monte Carlo Estimator

If the variance (univariate case for simplicity) of $f(x)$ satisfies

$$\sigma_f^2 \triangleq \mathbb{E}_{p(x)}\big(f^2(x)\big) - I^2(f) < \infty,$$

then the **variance of the estimator** is

$$\boxed{\operatorname{Var}\big(I_N(f)\big) = \frac{\sigma_f^2}{N}} \qquad \Longrightarrow \qquad \text{standard error} = \frac{\sigma_f}{\sqrt{N}}.$$

Proof: Unbiasedness and $\operatorname{Var}(I_N(f)) = \sigma_f^2/N$

Unbiasedness (scribe's addition): by linearity of expectation and $x^{(i)} \sim p$,

$$\mathbb{E}[I_N(f)] = \frac{1}{N}\sum_{i=1}^{N} \mathbb{E}_{p(x)}\big[f(x^{(i)})\big] = \frac{1}{N}\cdot N \cdot I(f) = I(f).$$

Variance. Recall $\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$, so

$$\operatorname{Var}(f) = \mathbb{E}_{p(x)}\big[(f - \mathbb{E}_{p(x)}[f])^2\big] = \mathbb{E}_{p(x)}[f^2] - \mathbb{E}_{p(x)}[f]^2 = \mathbb{E}_{p(x)}[f^2] - I(f)^2 = \sigma_f^2.$$

For i.i.d. X_n with variance σ^2, independence makes the covariances vanish, so

$$\operatorname{Var}(\overline{X}) = \operatorname{Var}\Big(\frac{1}{N}\sum_n X_n\Big) = \frac{1}{N^2}\sum_n \operatorname{Var}(X_n) = \frac{\sigma^2}{N}.$$

Letting $\overline{X} \coloneqq I_N(f)$ and $X \coloneqq f$:

$$\operatorname{Var}\big(I_N(f)\big) = \operatorname{Var}\Big(\frac{1}{N}\sum_n f_n\Big) = \frac{1}{N^2}\sum_n \operatorname{Var}(f_n) = \frac{\sigma_f^2}{N}. \qquad \square$$

Key Rule: Central Limit Theorem for the Monte Carlo Error

A **central limit theorem** yields **convergence in distribution** of the error:

$$\sqrt{N}\big(I_N(f) - I(f)\big) \underset{N\to\infty}{\Longrightarrow} \mathcal{N}\big(0, \sigma_f^2\big).$$

Mean and variance of the error (from the slides):

- **Expectation:** $\mathbb{E}[I_N(f) - I(f)] = 0$.
- **Variance:** $I(f)$ is a fixed number, so $\mathrm{Var}(I(f)) = 0$ and $\mathrm{Cov}(I_N(f), I(f)) = 0$. Hence

$$\mathrm{Var}\left[I_N(f) - I(f)\right] = \mathrm{Var}\left(I_N(f)\right) + \mathrm{Var}\left(I(f)\right) - 2\,\mathrm{Cov}\left(I_N(f), I(f)\right) = \mathrm{Var}\left(I_N(f)\right) = \frac{\sigma_f^2}{N}.$$

Multiplying by $\sqrt{N}$ gives an error with mean 0 and variance σ_f^2, matching the limit $\mathcal{N}(0, \sigma_f^2)$. *Scribe's notes:* (i) unbiasedness follows from linearity of expectation (above), not from the almost-sure convergence; a.s. convergence alone does not imply $\mathbb{E}[I_N] = I$. (ii) $\mathrm{Cov}(I_N(f), I(f)) = 0$ holds simply because $I(f)$ is a deterministic constant. (iii) The calculation fixes the mean and variance; that the limiting *shape* is Gaussian is the content of the classical CLT for i.i.d. sums with finite variance.

Key Rule: Monte Carlo for Optimization

The N samples can also be used to obtain a **maximum of the objective function** $p(x)$:

$$\hat{x} = \arg \max_{x^{(i)};\, i=1,\ldots,N} p\big(x^{(i)}\big).$$

Samples from p concentrate where p is large, so this is far cheaper than comparing all feasible solutions.

Key Difficulty: Why the $O(1/\sqrt{N})$ Rate Is Good News (Scribe's Note)

The rate $\sigma_f/\sqrt{N}$ **does not depend on the dimension** of $\mathcal{X}$. A deterministic grid with m points per axis needs m^D evaluations in D dimensions; Monte Carlo needs N samples whatever D is. The catch is that σ_f can be large, and – the subject of the rest of the lecture – we must be able to **draw samples from** p in the first place.

Worked Example: Estimating π (Scribe's Example)

Let p be uniform on the unit square $[0,1]^2$ and $f(x) = \mathbb{1}[x_1^2 + x_2^2 \le 1]$. Then $I(f) = \pi/4$ (a quarter disc), and since $f \in \{0,1\}$,

$$\sigma_f^2 = \mathbb{E}[f^2] - I(f)^2 = \tfrac{\pi}{4}\big(1 - \tfrac{\pi}{4}\big) \approx 0.7854 \times 0.2146 = 0.1685.$$

With $N = 10{,}000$ samples, the standard error of $I_N(f)$ is $\sqrt{0.1685/10^4} \approx 0.0041$, so $4I_N(f)$ estimates π with standard error ≈ 0.016. By the CLT, roughly 95% of runs land within ± 0.032 of π. To gain one more decimal digit we need $100\times$ more samples.

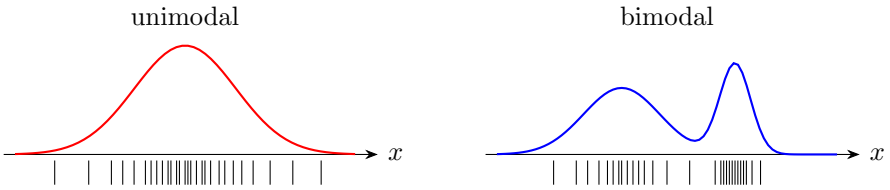
3.3 Non-Parametric Representation

Definition : Parametric vs. Non-Parametric Representation

Probability distributions can be represented:

1. **Parametrically**: e.g. by the mean and covariance of a Gaussian; or
2. **Non-parametrically**: by a set of *hypotheses* (samples) drawn from the distribution.

Advantage of the non-parametric representation: no restriction on the *type* of distribution (e.g. it can be multi-modal, non-Gaussian, etc.). **The more samples in an interval, the higher the probability of that interval.**



Densities (curves) and samples (ticks): sample density tracks probability. *(Schematic redrawn by the scribe.)*

Key Difficulty: How Do We Draw Samples?

The Monte Carlo principle assumes we can draw $x^{(i)} \sim p(x)$. For the distributions we care about – posteriors known only up to a normalizing constant – this is exactly what is hard. Sections 4–10 are different answers to **"how to draw samples from a function/distribution?"**

4 Rejection Sampling

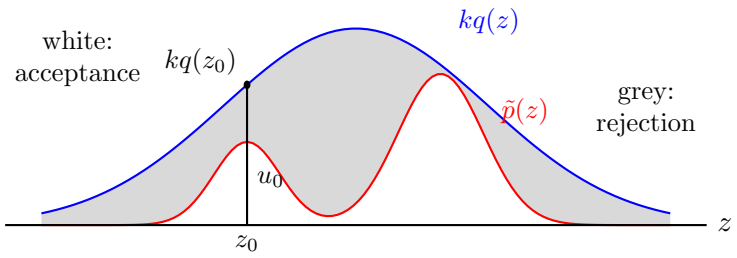
Definition : Setting for Rejection Sampling

- We wish to sample from a distribution $p(z)$ where **direct sampling is difficult**.
- Furthermore, we are **unable to easily evaluate** $p(z)$ because of an **unknown normalizing constant** Z_p:

$$p(z) = \frac{1}{Z_p}\tilde{p}(z),$$

  where $\tilde{p}(z)$ **can readily be evaluated**.
- We choose a **proposal distribution** $q(z)$ that is **easy to sample from** (e.g. a Gaussian), and a constant k such that

$$\tilde{p}(z) \leq k\, q(z) \qquad \text{for all values of } z.$$



Rejection sampling: draw $z_0 \sim q$, then a height u_0 uniform on $[0, kq(z_0)]$; accept if it falls under $\tilde{p}$. *(Redrawn after Bishop, Fig. 11.4.)*

Algorithm: Rejection Sampling

1: Set $i = 1$
2: **repeat** ▷ draw N samples
3: Sample $z^{(i)} \sim q(z)$ and $u \sim \mathcal{U}_{(0,1)}$ ▷ proposal and uniform
4: **if** $u < \dfrac{\tilde{p}(z^{(i)})}{k\, q(z^{(i)})}$ **then** ▷ u falls in the acceptance region
5: accept $z^{(i)}$ and increment i by 1
6: **else**
7: reject
8: **end if**
9: **until** $i = N$

Proof: Accepted Samples Are Distributed According to $p(z)$ (Scribe's Addition)

A proposal z is generated with density $q(z)$ and accepted with probability $\tilde{p}(z)/(k\, q(z)) \leq 1$. The joint density of "propose z and accept" is

$$q(z) \cdot \frac{\tilde{p}(z)}{k\, q(z)} = \frac{\tilde{p}(z)}{k}.$$

Integrating over z, the overall acceptance probability is

$$\Pr(\text{accept}) = \int \frac{\tilde{p}(z)}{k}\, dz = \frac{Z_p}{k}.$$

Conditioning on acceptance (Bayes' rule), the density of an accepted sample is

$$\frac{\tilde{p}(z)/k}{Z_p/k} = \frac{\tilde{p}(z)}{Z_p} = p(z). \qquad \blacksquare$$

Geometrically: (z_0, u_0) with $u_0 = u \cdot kq(z_0)$ is uniform under the curve $kq(z)$; keeping only points under $\tilde{p}(z)$ leaves points uniform under $\tilde{p}$, whose z-coordinates have density $\propto \tilde{p}(z)$.

Key Difficulty: Limitations of Rejection Sampling

- It is **not always possible** to bound $\tilde{p}(z)/q(z)$ with a reasonable constant k over the whole space $\mathcal{Z}$ (e.g. if q has lighter tails than p, no finite k exists).
- If k is **too large**, the acceptance probability

$$\Pr(z \text{ accepted}) = \Pr\left(u < \frac{\tilde{p}(z)}{kq(z)}\right) = \frac{1}{k}$$

  will be **too small**.
- This makes the method **impractical in high-dimensional spaces**.

Scribe's notes: (i) the slide's $1/k$ assumes $\tilde{p}$ is normalized ($Z_p = 1$); in general $\Pr(\text{accept}) = Z_p/k$ (proof above). (ii) Bishop's illustration of the curse of dimensionality: target $\mathcal{N}(\mathbf{0}, \sigma_p^2 I_D)$,

proposal $\mathcal{N}(\mathbf{0}, \sigma_q^2 I_D)$ with σ_q just 1% larger than σ_p. The smallest valid k is $(\sigma_q/\sigma_p)^D$; for $D = 1000$ this is $1.01^{1000} \approx 20{,}000$, so only about one proposal in 20,000 is accepted.

Worked Example: Acceptance Rate with an Unnormalized Target (Scribe's Example)

Target on $[0, 1]$: $\tilde{p}(z) = z(1-z)$, so $Z_p = \int_0^1 z(1-z)\,dz = \frac{1}{6}$ and $p(z) = 6z(1-z)$ (a Beta(2,2)). Proposal $q(z) = 1$ on $[0, 1]$. The smallest valid k is $\max_z \tilde{p}(z) = \frac{1}{4}$ (at $z = \frac{1}{2}$).

- Proposal $z = 0.5$: accept if $u < 0.25/0.25 = 1$ – always.
- Proposal $z = 0.1$: accept if $u < 0.09/0.25 = 0.36$.
- Overall acceptance rate: $Z_p/k = (1/6)/(1/4) = \frac{2}{3}$ (not $1/k = 4$, which is not even a probability – the reason for the Z_p correction).

5 Importance Sampling

Definition : Importance Sampling

- Given a **target distribution** $p(z)$ that is difficult to draw samples from directly.
- Importance sampling provides a framework for **approximating expectations** of a function $f(z)$ w.r.t. $p(z)$ – it does not produce samples from p itself.
- Samples $\{z^{(l)}\}$ are drawn from a simpler distribution $q(z)$, the **proposal distribution**.

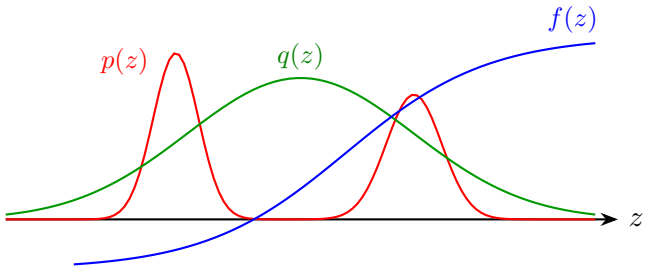
Key Rule: The Importance Sampling Estimator

Express the expectation as a finite sum over samples $\{z^{(l)}\}$ **weighted by the ratios** $p(z^{(l)})/q(z^{(l)})$:

$$\mathbb{E}[f] = \int f(\mathbf{z})\, p(\mathbf{z})\, d\mathbf{z} = \int f(\mathbf{z})\, \frac{p(\mathbf{z})}{q(\mathbf{z})}\, q(\mathbf{z})\, d\mathbf{z}$$
$$\simeq \frac{1}{L} \sum_{l=1}^{L} \frac{p(\mathbf{z}^{(l)})}{q(\mathbf{z}^{(l)})}\, f(\mathbf{z}^{(l)}), \qquad \mathbf{z}^{(l)} \sim q(\mathbf{z}).$$

This is the Monte Carlo estimator I_N of Section 3 applied to the function $f \cdot p/q$ under q.

- The quantities $r_l = p(\mathbf{z}^{(l)})/q(\mathbf{z}^{(l)})$ are the **importance weights**.
- They **correct the bias** introduced by sampling from the wrong distribution.
- Unlike rejection sampling, **all samples generated are retained**.
- *Scribe's note:* the identity requires $q(\mathbf{z}) > 0$ wherever $f(\mathbf{z})p(\mathbf{z}) \neq 0$; otherwise part of the integral is never sampled.



Importance sampling: samples from the broad q are reweighted to represent the target p. *(Redrawn after Bishop, Fig. 11.8.)*

5.1 Unnormalized Target and Proposal

Key Rule: Self-Normalized Importance Sampling

It is often the case that $\tilde{p}(z)$ **can be evaluated easily**, but not $p(z) = \tilde{p}(z)/Z_p$ (Z_p unknown). Define the proposal in a similar form, $q(z) = \tilde{q}(z)/Z_q$. Then

$$\mathbb{E}[f] = \int f(\mathbf{z})\, p(\mathbf{z})\, d\mathbf{z} = \frac{Z_q}{Z_p} \int f(\mathbf{z})\, \frac{\tilde{p}(\mathbf{z})}{\tilde{q}(\mathbf{z})}\, q(\mathbf{z})\, d\mathbf{z} \simeq \frac{Z_q}{Z_p}\, \frac{1}{L} \sum_{l=1}^{L} \tilde{r}_l\, f(\mathbf{z}^{(l)}), \qquad \tilde{r}_l = \frac{\tilde{p}(\mathbf{z}^{(l)})}{\tilde{q}(\mathbf{z}^{(l)})}.$$

The **same sample set** evaluates the ratio Z_p/Z_q:

$$\frac{Z_p}{Z_q} = \frac{1}{Z_q}\int \tilde{p}(\mathbf{z})\,d\mathbf{z} = \int \frac{\tilde{p}(\mathbf{z})}{\tilde{q}(\mathbf{z})}\,q(\mathbf{z})\,d\mathbf{z} \simeq \frac{1}{L}\sum_{l=1}^{L}\tilde{r}_l,$$

and hence

$$\boxed{\mathbb{E}[f] \simeq \sum_{l=1}^{L} w_l\, f(\mathbf{z}^{(l)}), \qquad w_l = \frac{\tilde{r}_l}{\sum_m \tilde{r}_m} = \frac{\tilde{p}(\mathbf{z}^{(l)})/\tilde{q}(\mathbf{z}^{(l)})}{\sum_m \tilde{p}(\mathbf{z}^{(m)})/\tilde{q}(\mathbf{z}^{(m)})}}$$

– normalized importance weights that are **easy to compute** (no Z_p or Z_q needed).

Proof: Self-Normalized Estimator

Substituting $\frac{Z_p}{Z_q} = \frac{1}{L}\sum_m \tilde{r}_m$ into $\mathbb{E}[f] = \frac{Z_q}{Z_p}\frac{1}{L}\sum_l \tilde{r}_l\, f(z^{(l)})$:

$$\mathbb{E}[f] = \frac{L}{\sum_m \tilde{r}_m}\,\frac{1}{L}\sum_l \tilde{r}_l\, f(z^{(l)}) = \sum_l \frac{\tilde{r}_l}{\sum_m \tilde{r}_m}\, f(z^{(l)}) = \sum_l w_l\, f(z^{(l)}). \qquad \square$$

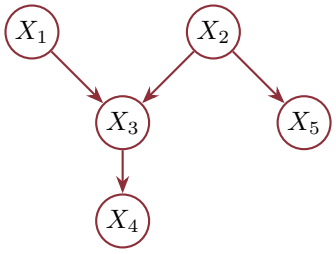
Scribe's note: because the same samples appear in numerator and denominator, the self-normalized estimator is **slightly biased** for finite L, but it is consistent (the bias vanishes as $L \to \infty$).

Key Difficulty: Limitations of Importance Sampling

- Success depends crucially on **how well** $q(z)$ **matches** $p(z)$.
- A strongly varying $p(z)f(z)$ has a significant proportion of its mass **concentrated over relatively small regions** of z space.
- The set of importance weights $\{r_l\}$ may then be **dominated by a few weights** with large values, the remaining weights being relatively insignificant. The estimate is effectively based on a handful of samples, and can be badly wrong with no warning (worse still if none of the samples lands in the high-mass region).

Scribe's note: a common diagnostic is the **effective sample size** $\text{ESS} = 1/\sum_l w_l^2$ (with normalized w_l): it equals L for equal weights and 1 if a single weight dominates.

5.2 Example on a Bayesian Network



	X_1=0	X_1=1
$p(x_1)$	0.6	0.4

	X_2=0	X_2=1
$p(x_2)$	0.7	0.3

$p(x_3 \mid x_1, x_2)$	X_3=0	X_3=1	X_3=2
X_1=0, X_2=0	0.3	0.4	0.3
X_1=0, X_2=1	0.05	0.25	0.7
X_1=1, X_2=0	0.9	0.08	0.02
X_1=1, X_2=1	0.5	0.3	0.2

$p(x_4 \mid x_3)$	X_4=0	X_4=1
X_3=0	0.1	0.9
X_3=1	0.4	0.6
X_3=2	0.99	0.01

$p(x_5 \mid x_2)$	X_5=0	X_5=1
X_2=0	0.95	0.05
X_2=1	0.2	0.8

X_1: Difficulty, X_2: Intelligence, X_3: Grade, X_4: Letter, X_5: SAT score (the student network of Koller & Friedman).

Worked Example: $p(x_1, x_4, x_5 \mid x_2 = 1, x_3 = 1)$ by Importance Sampling

Target. With query variables $x_F = \{x_1, x_4, x_5\}$ and evidence $x_E = \{x_2 = 1, x_3 = 1\}$,

$$p(x_1, x_4, x_5 \mid x_2 = 1, x_3 = 1) = \frac{p(x_1, x_4, x_5, x_2 = 1, x_3 = 1)}{p(x_2 = 1, x_3 = 1)} = \frac{p(x_F, x_E)}{p(x_E)} = \frac{1}{Z_p}\tilde{p}(x).$$

The numerator $\tilde{p}(x) = p(x_F, x_E)$ is a product of CPT entries (easy); we **don't want to evaluate** $Z_p = p(x_E)$, which needs a sum over all x_F.

Proposal. Clamp the evidence nodes (their CPTs become $p(x_2 = 1) = (0, 1)$ and $p(x_3 = 1 \mid x_1, x_2) = (0, 1, 0)$) and sample the rest from their CPTs:

$$\tilde{q}(x_1, x_4, x_5) = p(x_1)\, p(x_4 \mid x_3 = 1)\, p(x_5 \mid x_2 = 1),$$

i.e. $x_1 \sim p(x_1)$, $x_4 \sim p(x_4 \mid x_3 = 1)$, $x_5 \sim p(x_5 \mid x_2 = 1)$. Each is a discrete draw by a uniform number: e.g. $n = \text{rand}$; $x_1 = 0$ if $n < 0.6$, $x_1 = 1$ otherwise.

Weights. For the sample $x^{(l)} = \{x_1 = 0, x_4 = 1, x_5 = 1\}$:

$$\begin{aligned}
\tilde{p}(x^{(l)}) &= p(x_1{=}0)\, p(x_2{=}1)\, p(x_3{=}1 \mid x_2{=}1, x_1{=}0)\, p(x_4{=}1 \mid x_3{=}1)\, p(x_5{=}1 \mid x_2{=}1) \\
&= (0.6)(0.3)(0.25)(0.6)(0.8) = 0.0216, \\
\tilde{q}(x^{(l)}) &= p(x_1{=}0)\, p(x_4{=}1 \mid x_3{=}1)\, p(x_5{=}1 \mid x_2{=}1) = (0.6)(0.6)(0.8) = 0.288, \\
\Rightarrow\ \tilde{r}_l &= \frac{\tilde{p}(x^{(l)})}{\tilde{q}(x^{(l)})} = \frac{0.0216}{0.288} = 0.075.
\end{aligned}$$

Finally, the denominator $\sum_m \tilde{r}_m$ (hence each $w_l = \tilde{r}_l / \sum_m \tilde{r}_m$) is computed from all M samples.

Estimating the posterior. Sum the weights of the samples that match each configuration:

$$p(x_1{=}0, x_4{=}0, x_5{=}0 \mid x_2{=}1, x_3{=}1) \approx \frac{\sum_m w_m\, \delta\big(x^{(m)} = \{x_1{=}0, x_4{=}0, x_5{=}0\}\big)}{\sum_m w_m}, \quad \ldots$$

and in general

$$\boxed{p(x_F \mid x_E) \approx \frac{\sum_m w_m\, \delta(x^{(m)} = x_F)}{\sum_m w_m}}$$

where the denominator is a **normalizer** ensuring the probabilities sum to 1. (With normalized w_m, $\sum_m w_m = 1$ already.)

Worked Example: The Weights Are Evidence Likelihoods (Scribe's Example)

All CPT factors of sampled variables cancel between $\tilde{p}$ and $\tilde{q}$, leaving only the CPTs of the **evidence** nodes:

$$\tilde{r} = \frac{p(x_1)\,p(x_2{=}1)\,p(x_3{=}1 \mid x_1, x_2{=}1)\,p(x_4 \mid x_3{=}1)\,p(x_5 \mid x_2{=}1)}{p(x_1)\,p(x_4 \mid x_3{=}1)\,p(x_5 \mid x_2{=}1)} = p(x_2{=}1)\,p(x_3{=}1 \mid x_1, x_2{=}1).$$

So the weight depends only on x_1: $\tilde{r} = 0.3 \times 0.25 = 0.075$ if $x_1 = 0$ and $\tilde{r} = 0.3 \times 0.3 = 0.09$ if $x_1 = 1$. This scheme is known as **likelihood weighting**. With many samples, a fraction ≈ 0.6 have $x_1 = 0$, so

$$p(x_1{=}0 \mid x_E) \approx \frac{0.6 \times 0.075}{0.6 \times 0.075 + 0.4 \times 0.09} = \frac{0.045}{0.081} = 0.556,$$

which is the exact posterior (check by Bayes' rule). x_4 and x_5 are already drawn from their exact posteriors $p(x_4 \mid x_3{=}1)$ and $p(x_5 \mid x_2{=}1)$, since they are descendants of the evidence only.

6 Ancestral Sampling

Definition : Ancestral Sampling

Given a joint distribution that factorizes according to a DGM,

$$p(\mathbf{z}) = \prod_{i=1}^{M} p(\mathbf{z}_i \mid \mathrm{pa}_i),$$

where $\mathbf{z}_i$ is the set of variables associated with node i and pa_i the set of variables associated with the **parents** of node i:

- To obtain a sample from the joint, make **one pass** through the variables in the (topological) order $\mathbf{z}_1, \ldots, \mathbf{z}_M$, **sampling from the conditional distributions** $p(\mathbf{z}_i \mid \mathrm{pa}_i)$, with the parents already sampled.
- After one pass through the graph, we have obtained a sample from the joint distribution.

Worked Example: Ancestral Sampling on the Student Network

Question: how do we compute (sample from) $p(x_1, x_2, x_3, x_4, x_5)$?
Solution:

$$x_1 \sim p(x_1) = \bar{x}_1, \quad x_2 \sim p(x_2) = \bar{x}_2, \quad x_3 \sim p(x_3 \mid x_1 = \bar{x}_1, x_2 = \bar{x}_2) = \bar{x}_3,$$

$$x_4 \sim p(x_4 \mid x_3 = \bar{x}_3) = \bar{x}_4, \quad x_5 \sim p(x_5 \mid x_2 = \bar{x}_2) = \bar{x}_5.$$

One concrete pass (scribe's addition), sampling a discrete variable by comparing a uniform u with the cumulative CPT row:

- $u = 0.35 < 0.6 \Rightarrow \bar{x}_1 = 0$; $\quad u = 0.82 \geq 0.7 \Rightarrow \bar{x}_2 = 1$;
- row $(0.05, 0.25, 0.7)$ has cumulative $(0.05, 0.30, 1.0)$; $u = 0.41 \Rightarrow \bar{x}_3 = 2$;
- row $(0.99, 0.01)$; $u = 0.50 \Rightarrow \bar{x}_4 = 0$; $\quad$ row $(0.2, 0.8)$; $u = 0.13 \Rightarrow \bar{x}_5 = 0$.

The sample is $(0, 1, 2, 0, 0)$, which has joint probability $0.6 \cdot 0.3 \cdot 0.7 \cdot 0.99 \cdot 0.2 = 0.0249$. Repeating many passes, the **frequency** of each configuration estimates its joint probability.

Key Difficulty: Ancestral Sampling with Evidence (Scribe's Note)

Ancestral sampling draws from the **prior** joint. To condition on evidence at non-root nodes (e.g. $x_3 = 1$) one could discard every sample that disagrees with it ("logic sampling" – rejection sampling with $q =$ prior), but the acceptance rate is $p(x_E)$, which is tiny for unlikely evidence. Clamping the evidence and weighting the samples instead is exactly the **importance sampling** scheme of Section 5.

7 Sampling and the EM Algorithm

Key Rule: Monte Carlo EM

Sampling can **approximate the E-step** of EM (Lecture 7) for models in which the E-step **cannot be performed analytically**:

$$\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) = \int \underbrace{p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}})}_{\text{cannot be computed analytically}} \ln p(\mathbf{Z}, \mathbf{X} \mid \boldsymbol{\theta})\, d\mathbf{Z}.$$

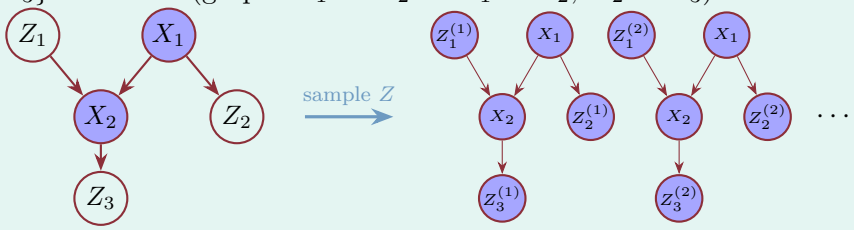
Approximate the integral by a **finite sum over samples** $\{\mathbf{Z}^{(l)}\}$ drawn from the current estimate $p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}})$:

$$\mathcal{Q}(\boldsymbol{\theta}, \boldsymbol{\theta}^{\text{old}}) \simeq \frac{1}{L} \sum_{l=1}^{L} \ln p\big(\mathbf{Z}^{(l)}, \mathbf{X} \mid \boldsymbol{\theta}\big).$$

- The samples $\mathbf{Z}^{(l)}$ are **treated as L i.i.d. observations**, so each term becomes a **complete-data log-likelihood**.
- The $\mathcal{Q}$ function is then **optimized in the usual way** in the M-step.
- This procedure is called the **Monte Carlo EM algorithm**.

Worked Example: Monte Carlo EM on a Partially Observed DGM

Goal: find the unknown parameters $\boldsymbol{\theta}$ of $p(X, Z \mid \boldsymbol{\theta})$, where $X = \{X_1, X_2\}$ are observed and $Z = \{Z_1, Z_2, Z_3\}$ are latent (graph: $Z_1 \to X_2 \leftarrow X_1 \to Z_2$, $X_2 \to Z_3$).



- **E-step:** sample $Z^{(l)} \sim p(Z \mid X, \boldsymbol{\theta}^{\text{old}})$ to obtain M **fully observed** i.i.d. Bayesian networks.
- **M-step:** follow Lecture 6 to find the **maximum complete likelihood**

$$\arg\max_{\boldsymbol{\theta}} \sum_{m} \sum_{n} \ln p\big(Z^{(m)}, X^{(n)} \mid \boldsymbol{\theta}\big),$$

  where the N i.i.d. observed X are **duplicated M times**. With complete data the MLE is closed-form: normalized counts of each CPT configuration.

Scribe's note on indexing: for each data case n we draw M completions $Z_n^{(m)} \sim p(Z_n \mid X^{(n)}, \boldsymbol{\theta}^{\text{old}})$, so the objective reads $\sum_n \sum_m \ln p(Z_n^{(m)}, X^{(n)} \mid \boldsymbol{\theta})$: every data case appears M times, each time with a different sampled completion of its latents. Dividing by M gives the $\frac{1}{L}\sum_l$ form above.

8 Markov Chain Monte Carlo (MCMC)

Definition : Markov Chain Monte Carlo

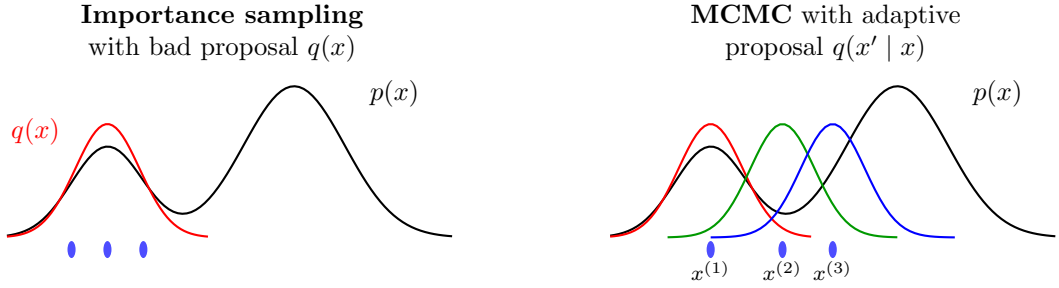
- MCMC is a strategy for generating samples $x^{(i)}$ while **exploring the state space** $\mathcal{X}$ using a **Markov chain**.
- The chain is constructed to **spend more time in the most important regions**.
- In particular, it is constructed so that the samples $x^{(i)}$ **mimic samples drawn from the target distribution** $p(x)$.
- *Scribe's note:* unlike Sections 4–6, successive MCMC samples are **not independent**; the law of large numbers still holds for ergodic chains, but the effective number of samples is smaller than N.

Key Rule: Adaptive Proposals

MCMC algorithms feature **adaptive proposals**:

- Instead of $q(x')$, we use $q(x' \mid x)$, where x' is the new state being sampled and x is the previous sample.
- As x changes, $q(x' \mid x)$ also changes (as a function of x').

With a fixed proposal, importance sampling can completely miss a mode of p that q does not cover; a proposal that moves with the chain can drift from one mode to the other.



Left: a fixed q never visits the right mode. Right: each proposal is centred at the current sample, so the chain walks across. *(Redrawn after the slides.)*

8.1 The Metropolis-Hastings Algorithm

Algorithm: Metropolis-Hastings

1: Initialize $x^{(0)}$
2: **for** $i = 0$ **to** $N - 1$ **do**
3: Sample $u \sim \mathcal{U}_{[0,1]}$ ▷ draw acceptance threshold
4: Sample $x' \sim q(x' \mid x^{(i)})$ ▷ draw from proposal
5: **if** $u < \mathcal{A}(x', x^{(i)}) = \min\left\{1, \dfrac{\tilde{p}(x')\, q(x^{(i)} \mid x')}{\tilde{p}(x^{(i)})\, q(x' \mid x^{(i)})}\right\}$ **then** ▷ acceptance probability
6: $x^{(i+1)} = x'$ ▷ new sample is accepted
7: **else**
8: $x^{(i+1)} = x^{(i)}$ ▷ rejected: duplicate the previous sample
9: **end if**

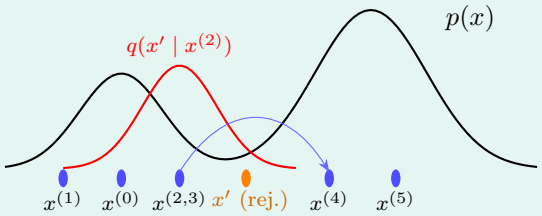
10: **end for**

Key Rule: Reading the Acceptance Probability

- Only the **ratio** $\tilde{p}(x')/\tilde{p}(x^{(i)})$ appears, so the unknown normalizer Z_p **cancels** – MH needs only $\tilde{p}$.
- Moves **uphill** in $\tilde{p}$ (with a symmetric proposal) are always accepted; moves downhill are accepted with probability equal to the ratio, so the chain does not just climb to a mode – it **samples**.
- The **Hastings correction** $q(x^{(i)} \mid x')/q(x' \mid x^{(i)})$ compensates for proposals that favour one direction.
- On rejection the **current state is repeated** in the output. These duplicates are essential: dropping them would bias the samples.

Worked Example: MH on a Bimodal Target

Goal: sample from a bimodal $p(x)$, with $q(x' \mid x)$ a Gaussian centred on x.



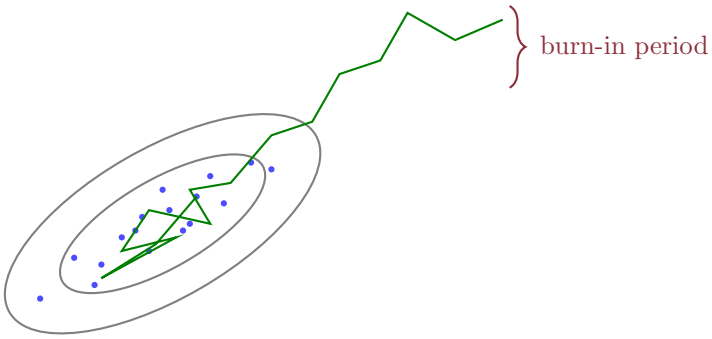
1. Initialize $x^{(0)}$ in the left mode.
2. Draw, accept $x^{(1)}$; draw, accept $x^{(2)}$ (small moves within the left mode).
3. Draw x' into the **valley** between the modes, and **reject**: set $x^{(3)} = x^{(2)}$.
4. Draw, accept $x^{(4)}$ (on the rising slope of the right mode); draw, accept $x^{(5)}$.

The adaptive proposal $q(x' \mid x^{(i)})$ allows us to **sample both modes** of $p(x)$!

Why x' was likely to be rejected (scribe's clarification): q is a symmetric Gaussian, so $q(x^{(2)} \mid x') = q(x' \mid x^{(2)})$ and the acceptance probability reduces to $\mathcal{A}(x', x^{(2)}) = \min\{1, \tilde{p}(x')/\tilde{p}(x^{(2)})\}$. Since x' lies in the low-density valley and $x^{(2)}$ near a mode, the ratio is small. Rejection is still a random event: with probability $\mathcal{A}$ the move would have been accepted.

Definition : Burn-In Period

- The **initial samples** may follow a very different distribution, especially if the starting point is in a region of **low density**.
- A **burn-in period** is therefore typically necessary: an initial number of samples (e.g. the first 1,000 or so) are **thrown away**.



The chain starts far out in a low-density region; its first steps are discarded. *(Redrawn after the UBC CPSC522 wiki figure.)*

Key Difficulty: Two Questions

What is the connection between Markov chains and MCMC? Why does the Metropolis-Hastings algorithm work? To answer them we need the theory of Markov chains: Sections 9 and 10.

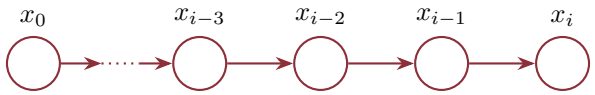
9 Markov Chains

Definition : Markov Chain on a Finite State Space

It is intuitive to introduce Markov chains on **finite state spaces**, where $x^{(i)}$ can only take s **discrete values**, $x^{(i)} \in \mathcal{X} = \{x_1, x_2, \ldots, x_s\}$. The stochastic process $x^{(i)}$ is called a **Markov chain** if

$$p\big(x^{(i)} \mid x^{(i-1)}, \ldots, x^{(1)}\big) = T\big(x^{(i)} \mid x^{(i-1)}\big),$$

where T is an $s \times s$ matrix. That is, the current state $x^{(i)}$ is **conditionally independent of all previous states given the most recent state** $x^{(i-1)}$.



This is the same first-order chain as the latent layer of the HMM (Lecture 8), with T playing the role of $\mathcal{A}$ – but now the chain is a **computational device** we design, not a model of data.

9.1 Properties of a Markov Chain

Definition : 1. Homogeneous Chain

The chain is **homogeneous** if $T \triangleq T(x^{(i)} \mid x^{(i-1)})$ **remains invariant** $\forall i$, with

$$\sum_{x^{(i)}} T\big(x^{(i)} \mid x^{(i-1)}\big) = 1 \quad \text{for any } i \qquad \text{(each row of } T \text{ sums to 1).}$$

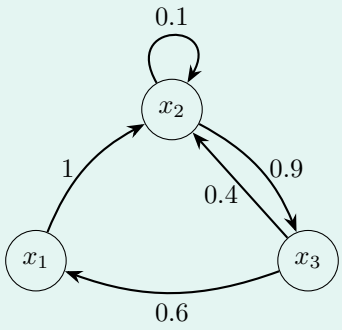
That is, the evolution of the chain in $\mathcal{X}$ depends solely on the current state and a **fixed transition matrix**. Convention: $T_{ab} = T(b \mid a)$, row = from, column = to; a distribution is a **row vector** μ, and one step maps $\mu \mapsto \mu T$.

Definition : 2. Stationary and Limiting Distributions

- A probability vector $\pi = p(x)$ on $\mathcal{X}$ is a **stationary (invariant) distribution** w.r.t. T if
$$\pi T = \pi.$$
- A **limiting distribution** π is a distribution over the states such that, **whatever the starting distribution** π_0, the Markov chain converges to π.

Scribe's note: $\pi T = \pi$ says π is a left eigenvector of T with eigenvalue 1. Every limiting distribution is stationary, but a stationary distribution need not be limiting (e.g. a chain that deterministically alternates between two states has stationary $\pi = (\frac{1}{2}, \frac{1}{2})$ but no limit).

Worked Example: Converging to the Stationary Distribution



$$T = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0.1 & 0.9 \\ 0.6 & 0.4 & 0 \end{bmatrix}$$

Transition graph for $\mathcal{X} = \{x_1, x_2, x_3\}$. *(Redrawn after Andrieu et al.)*

If the initial distribution is $\mu(x^{(1)}) = (0.5, 0.2, 0.3)$ (it can be any distribution):

$$\mu(x^{(2)}) = \mu(x^{(1)})T = (0.3 \cdot 0.6,\ \ 0.5 \cdot 1 + 0.2 \cdot 0.1 + 0.3 \cdot 0.4,\ \ 0.2 \cdot 0.9) = (0.18,\ 0.64,\ 0.18),$$

$$\vdots$$

$$\mu(x^{(t)}) = \mu(x^{(t-1)})T = (0.2213,\ 0.4098,\ 0.3689),$$

$$\mu(x^{(t+1)}) = \mu(x^{(t)})T = (0.2213,\ 0.4098,\ 0.3689).$$

The chain **converges to the stationary distribution**. (The slide's last line reads $\mu(x^{(t+1)}) = \mu(x^{(t+2)})T$; it should be $\mu(x^{(t)})T$.)

Exact check (scribe's addition). Solve $\pi T = \pi$: the first column gives $\pi_1 = 0.6\pi_3$; the third gives $\pi_3 = 0.9\pi_2$; so $\pi \propto (0.54,\ 1,\ 0.9)$, and normalizing by 2.44 gives $\pi = (0.2213,\ 0.4098,\ 0.3689)$ ✓.

Definition : 3. Irreducibility

A Markov chain is **irreducible** if from any state there is a **positive probability** of (eventually) visiting every other state:

$$\forall a, b \in \mathcal{X},\ \exists\, t \geq 0 \quad \text{s.t.} \quad p(x_t = b \mid x_0 = a) > 0.$$

Definition : 4. Aperiodicity

The Markov chain should **not get trapped in cycles**:

$$\gcd\{t \geq 1 :\ p(x_t = a \mid x_0 = a) > 0\} = 1, \qquad \forall a \in \mathcal{X},$$

where gcd is the **greatest common divisor**. *Example:* in the chain above, x_2 has a self-loop (return time 1), so gcd = 1 for x_2; and since the chain is irreducible, all states share the same period, so the chain is aperiodic.

Key Rule: Ergodic Theorem for Markov Chains

- A Markov chain is **ergodic** if it is **irreducible and aperiodic**.
- **Ergodicity is important**: it implies we can reach the stationary/limiting distribution

π, **no matter the initial distribution** π_0 (and, on a finite space, that π is unique).
- All good MCMC algorithms **must satisfy ergodicity**, so that we cannot initialize in a way that will never converge.

9.2 Detailed Balance

Definition : Detailed Balance (Reversibility)

A probability vector $\pi = p(x)$ on $\mathcal{X}$ **satisfies detailed balance** w.r.t. T if

$$\pi_a T_{ab} = \pi_b T_{ba}, \qquad \forall a, b \in \mathcal{X},$$

where $T_{ab} = T(b \mid a)$ is the probability of moving from a to b; each row of T sums to 1. In words: in equilibrium, the **probability flow** from a to b equals the flow from b to a, for every pair.

Proof: Remark 1 – Detailed Balance $\Longrightarrow$ Stationarity ($\pi T = \pi$)

Show that every component of the row vector πT equals the matching component of π. For any state $b \in \mathcal{X}$:

$$\begin{aligned}
(\pi T)_b &= \textstyle\sum_a \pi_a T_{ab} && \text{(definition: } b\text{-th entry of } \pi T\text{)} \\
&= \textstyle\sum_a \pi_b T_{ba} && \text{(detailed balance, term by term)} \\
&= \pi_b \textstyle\sum_a T_{ba} && (\pi_b \text{ does not depend on } a) \\
&= \pi_b. && \text{(row } b \text{ of } T \text{ sums to 1)}
\end{aligned}$$

This holds for every $b \in \mathcal{X} \Longrightarrow$ all components agree $\Longrightarrow \pi T = \pi$. ■

Key Rule: Remark 2 – Detailed Balance = "Reversibility"

- Just **terminology**: a Markov chain is called "reversible" if it has a stationary distribution π that satisfies detailed balance w.r.t. T (run in equilibrium, it looks statistically the same forwards and backwards in time).
- The **converse of Remark 1 is not true**: a stationary distribution need not satisfy detailed balance. Detailed balance is a **sufficient** condition, which is why it is **convenient to check** – it involves only pairs of states, with no sum over $\mathcal{X}$.

Worked Example: A Stationary Chain That Is Not Reversible (Scribe's Example)

The 3-state chain above has stationary $\pi = (0.2213, 0.4098, 0.3689)$, but

$$\pi_1 T_{12} = 0.2213 \times 1 = 0.2213 \;\neq\; \pi_2 T_{21} = 0.4098 \times 0 = 0.$$

Probability circulates around the cycle $x_1 \to x_2 \to x_3 \to x_1$ instead of balancing pairwise, so detailed balance fails although $\pi T = \pi$ – a concrete instance of Remark 2.

10 Why Does Metropolis-Hastings Work?

Definition : Goal and Setup

Goal: show that the target p is **stationary** for the MH chain, and state the **conditions for convergence** to p.
Setup: target $p(x) = \tilde{p}(x)/Z$ with $0 < Z < \infty$. Propose $x' \sim q(\cdot \mid x)$ and accept with probability

$$A(x', x) = \min\left\{1, \frac{\tilde{p}(x')\, q(x \mid x')}{\tilde{p}(x)\, q(x' \mid x)}\right\}.$$

For a discrete state space, the **full transition rule** is

$$\begin{aligned} T(x' \mid x) &= q(x' \mid x)\, A(x', x), && x' \neq x, \\ T(x \mid x) &= 1 - \sum_{y \neq x} q(y \mid x)\, A(y, x). \end{aligned}$$

Rejection leaves the chain at its current state x: the self-transition collects the proposal $x' = x$ (if possible) plus all the rejected mass.
Strategy: show detailed balance for the MH chain,

$$p(x)\, T(x' \mid x) = p(x')\, T(x \mid x'), \qquad \text{for all } x, x',$$

then Remark 1 gives stationarity.

Proof: The MH Chain Satisfies Detailed Balance

For distinct states $x \neq x'$, define the **proposal flows**

$$a = p(x)\, q(x' \mid x), \qquad b = p(x')\, q(x \mid x').$$

Z **cancels** in the acceptance ratio, since $\tilde{p} = Zp$:

$$\frac{\tilde{p}(x')\, q(x \mid x')}{\tilde{p}(x)\, q(x' \mid x)} = \frac{b}{a}, \qquad \text{so} \qquad A(x', x) = \min\{1,\ b/a\}, \quad A(x, x') = \min\{1,\ a/b\}.$$

When $a > 0$ **and** $b > 0$**:**

$$\begin{aligned} p(x)\, T(x' \mid x) &= p(x)\, q(x' \mid x)\, A(x', x) = a \min\{1,\ b/a\} = \min\{a,\ b\}, \\ p(x')\, T(x \mid x') &= p(x')\, q(x \mid x')\, A(x, x') = b \min\{1,\ a/b\} = \min\{a,\ b\}. \end{aligned}$$

$\Longrightarrow$ both directions have the **same accepted probability flow**.
Remaining cases:

(i) If $a = 0$ or $b = 0$, both accepted flows are zero (the side with zero proposal flow is trivially zero; on the other side, the acceptance ratio is 0, taking the convention $A = 0$ when $\tilde{p}(x') = 0$).
(ii) For $x' = x$, detailed balance $p(x)T(x \mid x) = p(x)T(x \mid x)$ holds trivially.

From detailed balance to stationarity. Summing $p(x)T(x' \mid x) = p(x')T(x \mid x')$ over all starting states x:

$$\sum_x p(x)\, T(x' \mid x) = p(x') \sum_x T(x \mid x') = p(x').$$

The sum $\sum_x T(x \mid x') = 1$ because, from x', the chain must move to **some** state. Therefore $pT = p$ (as in Remark 1): **the normalized target p is stationary!** ■

Key Rule: From Stationarity to Convergence

Detailed balance establishes that p is stationary, but on its own it does not guarantee convergence to p! For a finite state space, suppose the MH chain is also:

- **Irreducible:** every state with $p(x) > 0$ can reach every other such state in finitely many steps with positive probability;
- **Aperiodic:** $\gcd\{t \geq 1 : p(x_t = a \mid x_0 = a) > 0\} = 1$, $\forall a \in \mathcal{X}$. For an irreducible chain, a **self-loop** at one state, $p(x_1 = a \mid x_0 = a) = T(a \mid a) > 0$ for some a, suffices; **MH has this whenever a proposal can be rejected**.

Then p is the **unique** stationary distribution and

$$\mu_0 T^n \to p \quad \text{as } n \to \infty, \qquad \text{for every initial distribution } \mu_0.$$

In practice, choose q so that every state with $p(x) > 0$ is reachable, e.g. $q(x' \mid x) > 0$ on the support of p.

Worked Example: Building the MH Transition Matrix by Hand (Scribe's Example)

Target on three states: $\tilde{p} = (1, 2, 3)$, so $p = (\frac{1}{6}, \frac{2}{6}, \frac{3}{6})$ (MH never needs $Z = 6$). Proposal: pick one of the **other** two states uniformly, $q(y \mid x) = \frac{1}{2}$ for $y \neq x$ – symmetric, so $A(y, x) = \min\{1, \tilde{p}(y)/\tilde{p}(x)\}$.

$$T = \begin{bmatrix} 0 & \frac{1}{2} \cdot 1 & \frac{1}{2} \cdot 1 \\ \frac{1}{2} \cdot \frac{1}{2} & \frac{1}{4} & \frac{1}{2} \cdot 1 \\ \frac{1}{2} \cdot \frac{1}{3} & \frac{1}{2} \cdot \frac{2}{3} & \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.25 & 0.25 & 0.5 \\ \frac{1}{6} & \frac{1}{3} & 0.5 \end{bmatrix}.$$

The diagonal is 1 minus the off-diagonal row sum: from state 2, the proposal to state 1 is rejected half the time, leaving $T(2 \mid 2) = \frac{1}{4}$.

Detailed balance: $p_1 T_{12} = \frac{1}{6} \cdot \frac{1}{2} = \frac{1}{12} = \frac{2}{6} \cdot \frac{1}{4} = p_2 T_{21}$; $p_1 T_{13} = \frac{1}{12} = \frac{3}{6} \cdot \frac{1}{6} = p_3 T_{31}$; $p_2 T_{23} = \frac{1}{6} = \frac{3}{6} \cdot \frac{1}{3} = p_3 T_{32}$ ✓.

Stationarity: $pT = \left(0 + \frac{1}{12} + \frac{1}{12},\ \frac{1}{12} + \frac{1}{12} + \frac{1}{6},\ \frac{1}{12} + \frac{1}{6} + \frac{1}{4}\right) = (\frac{1}{6}, \frac{1}{3}, \frac{1}{2}) = p$ ✓.

Convergence: all off-diagonal proposals are positive (irreducible), and $T(2 \mid 2), T(3 \mid 3) > 0$ (aperiodic), so the chain converges to p from any start.

11 Special Cases of Metropolis-Hastings

11.1 The Metropolis Algorithm

Key Rule: Metropolis Algorithm

- The Metropolis algorithm is a **special case** of the Metropolis-Hastings algorithm.
- The proposal distribution is a **random walk**, i.e. symmetric, $q(x \mid x') = q(x' \mid x)$ – e.g. an isotropic Gaussian.
- The acceptance probability simplifies because the proposal terms **cancel**:

$$\mathcal{A}(x', x) = \min\left\{1,\ \frac{\tilde{p}(x')\,\cancel{q(x \mid x')}}{\tilde{p}(x)\,\cancel{q(x' \mid x)}}\right\} = \min\left\{1,\ \frac{\tilde{p}(x')}{\tilde{p}(x)}\right\}.$$

Worked Example: Metropolis on a Correlated 2D Gaussian

Bishop's illustration (Fig. 11.9): sampling a correlated Gaussian (the ellipse shows its one-standard-deviation contour) with an isotropic Gaussian proposal. Of **150 candidate samples**, **43 are rejected** (green steps: accepted; red: rejected), an acceptance rate of $107/150 \approx 71\%$. *Scribe's note – the step-size trade-off:* a small proposal variance gives high acceptance but tiny steps (slow random-walk exploration of the long axis); a large variance proposes many points off the ellipse and gets rejected often. The proposal scale must be tuned to the narrowest direction of p, which is what makes random-walk Metropolis slow for strongly correlated targets.

11.2 Gibbs Sampling

Definition : Gibbs Sampling

Suppose we have an N-dimensional vector x and expressions for the **full conditionals**

$$p(x_j \mid x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_N).$$

For $j = 1, \ldots, N$ we use the proposal distribution

$$q\big(x' \mid x^{(i)}\big) = \begin{cases} p\big(x'_j \mid x^{(i)}_{\setminus j}\big) & \text{if } x'_{\setminus j} = x^{(i)}_{\setminus j}, \\ 0 & \text{otherwise,} \end{cases}$$

i.e. resample **only component** j from its full conditional, keeping all other components fixed ($x_{\setminus j}$ denotes all components except x_j).

Proof: Gibbs Sampling Is MH with Acceptance Probability Always 1

Write $p(x) = p(x_j \mid x_{\setminus j})\, p(x_{\setminus j})$ and $p(x') = p(x'_j \mid x'_{\setminus j})\, p(x'_{\setminus j})$. Then

$$\mathcal{A}(x', x) = \min\left\{1,\ \frac{p(x')\, q(x \mid x')}{p(x)\, q(x' \mid x)}\right\} = \min\left\{1,\ \frac{p(x'_j \mid x'_{\setminus j})\, p(x'_{\setminus j})\, p(x_j \mid x'_{\setminus j})}{p(x_j \mid x_{\setminus j})\, p(x_{\setminus j})\, p(x'_j \mid x_{\setminus j})}\right\}.$$

We use $x'_{\setminus j} = x_{\setminus j}$ because these components are **kept fixed** during the sampling step:

$$\implies \mathcal{A}(x', x) = \min\left\{1, \frac{\cancel{p(x'_j \mid x_{\setminus j})}\, \cancel{p(x_{\setminus j})}\, \cancel{p(x_j \mid x_{\setminus j})}}{\cancel{p(x_j \mid x_{\setminus j})}\, \cancel{p(x_{\setminus j})}\, \cancel{p(x'_j \mid x_{\setminus j})}}\right\} = \min\{1, 1\} = 1. \qquad \square$$

Every Gibbs proposal is accepted, so no $u \sim \mathcal{U}$ is drawn.

Algorithm: Gibbs Sampling

Initialize $\{x_i : i = 1, \ldots, M\}$
for $\tau = 1, \ldots, T$ **do**
  Sample $x_1^{(\tau+1)} \sim p\big(x_1 \mid x_2^{(\tau)}, x_3^{(\tau)}, \ldots, x_M^{(\tau)}\big)$
  Sample $x_2^{(\tau+1)} \sim p\big(x_2 \mid x_1^{(\tau+1)}, x_3^{(\tau)}, \ldots, x_M^{(\tau)}\big)$
  $\vdots$
  Sample $x_j^{(\tau+1)} \sim p\big(x_j \mid x_1^{(\tau+1)}, \ldots, x_{j-1}^{(\tau+1)}, x_{j+1}^{(\tau)}, \ldots, x_M^{(\tau)}\big)$
  $\vdots$
  Sample $x_M^{(\tau+1)} \sim p\big(x_M \mid x_1^{(\tau+1)}, x_2^{(\tau+1)}, \ldots, x_{M-1}^{(\tau+1)}\big)$
end for

Each update conditions on the **most recent** values: components already updated in this sweep use $\tau + 1$, the rest τ. (Named after Josiah Willard Gibbs, 1839–1903.)

Key Rule: Full Conditionals Reduce to the Markov Blanket

The conditional $p(x_j \mid x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_N)$ looks intimidating, but recall **Markov blankets**:

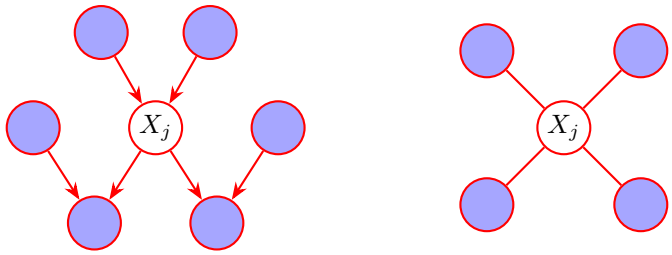
$$p(x_j \mid x_1, \ldots, x_{j-1}, x_{j+1}, \ldots, x_N) = p\big(x_j \mid \mathrm{MB}(x_j)\big).$$

- **Bayesian network:** the Markov blanket of X_j is its **parents, children, and co-parents**. Concretely,

$$p(x_j \mid x_{\setminus j}) \;\propto\; p(x_j \mid \mathrm{pa}_j) \prod_{c \in \mathrm{ch}(j)} p(x_c \mid \mathrm{pa}_c)$$

  – only the CPTs that mention x_j survive; all others cancel in the normalization.
- **MRF:** the Markov blanket of X_j is its **immediate neighbours**, and $p(x_j \mid x_{\setminus j}) \propto \prod_{C \ni j} \psi_C(x_C)$.

Each Gibbs step is therefore a **local** computation: a K-way discrete distribution for discrete variables.

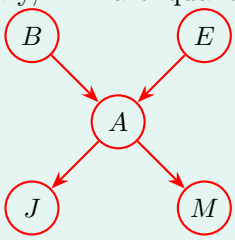


Bayesian network MRF

Markov blankets (shaded). *(Redrawn after Bishop.)*

Worked Example: Gibbs Sampling on the Burglary Network

B: Burglary, E: Earthquake, A: Alarm, J: John calls, M: Mary calls.



$p(B) = 0.001, \;\; p(E) = 0.002$

B	E	$p(A \mid B, E)$
T	T	0.95
T	F	0.94
F	T	0.29
F	F	0.01

A	$p(J \mid A)$	$p(M \mid A)$
T	0.90	0.70
F	0.05	0.01

Sample in the order B, E, A, J, M; initialize all variables at $t = 0$ to False. The Markov blankets are $\text{MB}(B) = \{A, E\}$, $\text{MB}(E) = \{A, B\}$, $\text{MB}(A) = \{B, E, J, M\}$, $\text{MB}(J) = \{A\}$, $\text{MB}(M) = \{A\}$.

Sweep $t = 1$ (the normalized probabilities are the scribe's addition):

- $p(B \mid A, E) \propto p(A \mid B, E)\, p(B)$ with $(A, E) = (F, F)$:

$$p(B{=}T \mid \cdot) \propto (0.06)(0.001) = 0.00006, \quad p(B{=}F \mid \cdot) \propto (0.99)(0.999) = 0.98901.$$

  Normalized: $p(B{=}T \mid \cdot) \approx 6 \times 10^{-5}$. Sample $B = F$.
- $p(E \mid A, B) \propto p(A \mid B, E)\, p(E)$ with $(A, B) = (F, F)$:

$$p(E{=}T \mid \cdot) \propto (0.71)(0.002) = 0.00142, \quad p(E{=}F \mid \cdot) \propto (0.99)(0.998) = 0.98802.$$

  Normalized: $p(E{=}T \mid \cdot) \approx 0.0014$. Sample $E = T$.
- $p(A \mid B, E, J, M) \propto p(J \mid A)\, p(M \mid A)\, p(A \mid B, E)$ with $(B, E, J, M) = (F, T, F, F)$:

$$p(A{=}T \mid \cdot) \propto (0.1)(0.3)(0.29) = 0.0087, \quad p(A{=}F \mid \cdot) \propto (0.95)(0.99)(0.71) = 0.6678.$$

  Normalized: $p(A{=}T \mid \cdot) \approx 0.013$. Sample $A = F$.
- $p(J \mid A)$ with $A = F$: $p(J{=}T \mid A{=}F) = 0.05$, $p(J{=}F \mid A{=}F) = 0.95$. Sample $J = T$.
- $p(M \mid A)$ with $A = F$: $p(M{=}T \mid A{=}F) = 0.01$, $p(M{=}F \mid A{=}F) = 0.99$. Sample $M = F$.

Later sweeps repeat the procedure for $t = 2, 3, 4, \ldots$:

t	B	E	A	J	M
0	F	F	F	F	F
1	F	T	F	T	F
2	F	T	T	T	T
3	T	F	T	F	T
4	T	F	T	F	F

Scribe's note: the draws $E = T$ and $J = T$ at $t = 1$ have probabilities of only about 0.0014 and 0.05; the slide chooses them (and the rows for $t = 2$–4) to make the illustration move. A real run from this initialization would stay at all-False for a very long time.

Worked Example: Inference from the Gibbs Samples

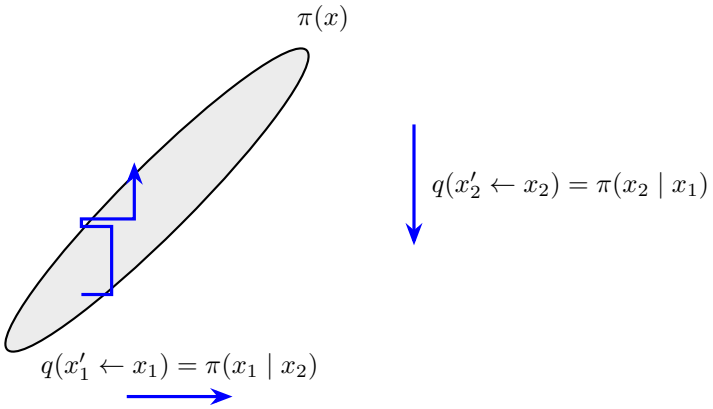
Question: taking the initialization and ignoring burn-in, estimate $p(B \mid E)$ and $p(A)$ from the 5 samples above.
Solution: count frequencies.

- $p(B \mid E)$: the rows with $E = F$ are $t = 0, 3, 4$, with $B = F, T, T$; the rows with $E = T$ are $t = 1, 2$, both with $B = F$:

  $$p(B{=}F \mid E{=}F) = \tfrac{1}{3}, \quad p(B{=}T \mid E{=}F) = \tfrac{2}{3}; \quad p(B{=}F \mid E{=}T) = 1, \quad p(B{=}T \mid E{=}T) = 0.$$

- $p(A)$: $A = T$ at $t = 2, 3, 4$, so $p(A{=}T) = \frac{3}{5}$, $p(A{=}F) = \frac{2}{5}$.

Scribe's note: these numbers illustrate the **counting procedure**, not good estimates – the true values are $p(B{=}T \mid E) = 0.001$ (since $B \perp\!\!\!\perp E$ a priori) and $p(A{=}T) = \sum_{B,E} p(A{=}T \mid B, E)p(B)p(E) \approx 0.0115$. Five correlated samples with no burn-in cannot approach them; in practice one discards a burn-in and averages over many thousands of sweeps.



Gibbs sampling moves along one coordinate axis at a time. *(Redrawn after the slideplayer figure.)*

Key Difficulty: When Gibbs Sampling Struggles (Scribe's Note)

- **Strong correlation:** as the figure shows, axis-aligned moves on a narrow diagonal ellipse are tiny, so the chain mixes slowly (the same issue as random-walk Metropolis).
- **Zero probabilities:** if some full conditionals are deterministic (e.g. a CPT entry of 0 or 1), single-site updates may be unable to leave a region, and the chain is **not irreducible** – convergence to p is then not guaranteed.
- **Validity:** each single-site update is an MH step with acceptance 1, hence leaves p invariant; a full sweep is a composition of such steps and also leaves p invariant.

12 Sampling from Continuous Distributions

Key Rule: Inverse-CDF (Inverse Transform) Sampling

1. Convert the **probability density function** (PDF) $f(x)$ into the **cumulative distribution function** (CDF):

$$y = F(x) = \int_{-\infty}^{x} f(t)\,dt, \qquad 0 \le y \le 1.$$

2. Draw a sample from the **uniform distribution**: $u \sim \text{Unif}(0,1)$.
3. The sample from $f(x)$ is $x = F^{-1}(u)$.

Proof: $F^{-1}(U)$ Has CDF F (Scribe's Addition)

Let $U \sim \text{Unif}(0,1)$ and $X = F^{-1}(U)$, with F continuous and strictly increasing. For any x,

$$\Pr(X \le x) = \Pr\left(F^{-1}(U) \le x\right) = \Pr\left(U \le F(x)\right) = F(x),$$

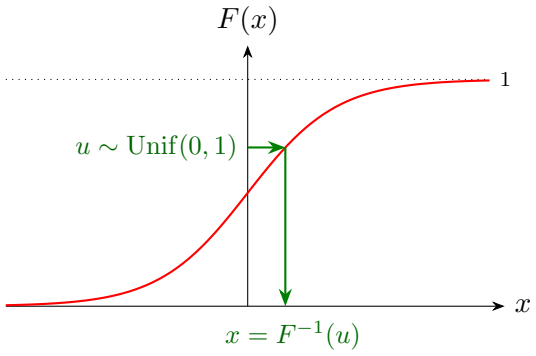
because F is increasing and $\Pr(U \le y) = y$ for $y \in [0,1]$. So X has CDF F, i.e. density f. □
The discrete rule used in Sections 5–6 ("$x_1 = 0$ if $u < 0.6$, else $x_1 = 1$") is the same method with a step-function CDF.

Worked Example: Sampling a Univariate Gaussian

PDF: $f(x) = \dfrac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\dfrac{(x-\mu)^2}{2\sigma^2}\right\}$.
CDF: $F(x) = \displaystyle\int_{-\infty}^{x} f(t)\,dt = \frac{1}{2}\left[1 + \text{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$, where $\text{erf}(x) = \dfrac{2}{\sqrt{\pi}} \displaystyle\int_0^x \exp\{-t^2\}\,dt$.
Sampling: with $u \sim \text{Unif}(0,1) = 0.5$:

- for $\mu = 0$ (any σ^2), $x = F^{-1}(0.5) = 0$;
- for $\mu = -2$, $\sigma^2 = 0.5$, $x = F^{-1}(0.5) = -2$.

$u = 0.5$ always returns the **median**, which for a Gaussian is μ. (On the slide the second label reads $F^{-1}(x)$; it should be $F^{-1}(u)$.) In general $x = \mu + \sigma\sqrt{2}\,\text{erf}^{-1}(2u-1)$; e.g. $u = 0.8413$ gives $x = \mu + \sigma$.



Inverse-CDF sampling: a uniform height on the F-axis is mapped back through the CDF. *(Schematic by the scribe; standard normal CDF.)*

Key Difficulty: Limits of the Inverse CDF (Scribe's Note)

The method needs F in closed form and invertible. That works for many univariate distributions (exponential: $x = -\ln(1-u)/\lambda$), but not for high-dimensional or unnormalized targets – which is exactly why the rejection, importance and MCMC methods of this lecture exist. In practice, Gaussian samples are generated by specialized transforms (e.g. Box-Muller) rather than by inverting erf.

13 Summary

- Explained the **Monte Carlo principle** and its justification for sampling methods.
- Applied **Rejection**, **Importance**, **Ancestral**, **Metropolis-Hastings**, **Metropolis** and **Gibbs** sampling methods to do maximal probability, approximate inference, and expectation.
- Used **Markov chain properties**, i.e. homogeneous, stationary distribution, irreducibility, aperiodicity, ergodicity and detailed balance, to show validity of the MH algorithm.

Summary: Comparison of Sampling Methods

Method	Needs	Samples	Output	Main weakness
Rejection	$\tilde{p}$, q, bound k	i.i.d., exact	samples from p	acceptance Z_p/k vanishes in high D
Importance	$\tilde{p}$, q ($q>0$ where $fp \neq 0$)	i.i.d., weighted	$\mathbb{E}_p[f]$ via $\sum_l w_l f(z^{(l)})$	weights dominated by few samples
Ancestral	DGM with CPTs	i.i.d., exact	samples from the joint	evidence must be handled separately
Metropolis-Hastings	$\tilde{p}$, $q(x' \mid x)$	correlated chain	samples from p after burn-in	tuning q; slow mixing
Metropolis	$\tilde{p}$, symmetric q	correlated chain	as MH, $\mathcal{A} = \min\{1, \tilde{p}(x')/\tilde{p}(x)\}$	random-walk behaviour
Gibbs	full conditionals $p(x_j \mid \mathrm{MB}(x_j))$	correlated chain	as MH, $\mathcal{A} = 1$	strong correlations; zeros

Key Takeaways

1. Normalization, marginalization and expectation are high-dimensional integrals; **Monte Carlo** replaces them by sample averages $I_N(f) = \frac{1}{N}\sum_i f(x^{(i)})$, which are **unbiased**, converge **almost surely** (strong LLN), and have variance σ_f^2/N with a Gaussian error (CLT) – a rate independent of dimension.
2. **Rejection sampling** accepts $z \sim q$ with probability $\tilde{p}(z)/(kq(z))$, given $\tilde{p} \leq kq$; accepted samples are exact draws from p, but the acceptance rate collapses in high dimensions.
3. **Importance sampling** keeps every sample and reweights it: $\mathbb{E}[f] \simeq \sum_l w_l f(z^{(l)})$ with $w_l = \tilde{r}_l / \sum_m \tilde{r}_m$, $\tilde{r}_l = \tilde{p}(z^{(l)})/\tilde{q}(z^{(l)})$. On a BN with clamped evidence, the weights are the evidence CPT entries (likelihood weighting). It fails when q matches p poorly.
4. **Ancestral sampling** draws from a DGM's joint by sampling each node given its already-sampled parents in topological order. **Monte Carlo EM** replaces the E-step integral by

$\frac{1}{L}\sum_l \ln p(\mathbf{Z}^{(l)}, \mathbf{X} \mid \boldsymbol{\theta})$ with $\mathbf{Z}^{(l)} \sim p(\mathbf{Z} \mid \mathbf{X}, \boldsymbol{\theta}^{\text{old}})$.

5. **Metropolis-Hastings** proposes $x' \sim q(x' \mid x)$ and accepts with $\mathcal{A} = \min\{1, \tilde{p}(x')q(x \mid x')/[\tilde{p}(x)q(x' \mid x)]\}$; rejection repeats the current state. Discard a **burn-in**.
6. MH works because its kernel satisfies **detailed balance** $p(x)T(x' \mid x) = p(x')T(x \mid x')$ (both sides equal $\min\{a, b\}$), which implies **stationarity** $pT = p$. Stationarity alone is not enough: **irreducibility** and **aperiodicity** (ergodicity) give convergence $\mu_0 T^n \to p$ from any start. Detailed balance is sufficient, not necessary, for stationarity.
7. **Metropolis**: symmetric q, so $\mathcal{A} = \min\{1, \tilde{p}(x')/\tilde{p}(x)\}$. **Gibbs**: propose each x_j from its full conditional $p(x_j \mid \text{MB}(x_j))$; the MH acceptance is always 1.
8. **Inverse-CDF sampling**: $x = F^{-1}(u)$, $u \sim \text{Unif}(0, 1)$, draws exactly from a univariate distribution with an invertible CDF.

Acknowledgement: Slide content adapted from C. Andrieu, N. de Freitas, A. Doucet and M. I. Jordan, "An Introduction to MCMC for Machine Learning"; C. Bishop, "Pattern Recognition and Machine Learning" (Ch. 11); E. Xing, CMU 10-708 lectures 16–17; K. Murphy, "Machine Learning: A Probabilistic Perspective" (Ch. 23–24); D. Koller and N. Friedman, "Probabilistic Graphical Models" (Ch. 12).